

Vamshi Krishna Reddy Rajanala

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Professional Summary

Electronic engineer (MS, 4.0/4.0) with hands-on schematic capture, PCB layout review, and board-level bring-up experience across analog, digital, and RF systems. Validates hardware functionality with oscilloscopes, multimeters, and logic analyzers, and debugs board-level issues to root cause. Familiar with AI/GPU server and high-performance compute rack architectures through coursework and independent study of platforms comparable to the GB3600 ecosystem. Documents engineering work clearly and collaborates across hardware, firmware, and software teams.

Education

Texas A&M University, Kingsville

Master of Science, Electrical Engineering, GPA: 4.0 / 4.0 | Aug 2024 - May 2026

Jawaharlal Nehru Technological University, Hyderabad

Bachelor of Technology, Electronics and Communication Engineering, GPA: 3.25 / 4.0 | Jul 2019 - Jun 2023

Relevant Coursework: Digital System Design, VLSI Design, System on Chip Architecture, Electronic Devices and Circuits, Network Analysis and Transmission Lines, Electromagnetic Fields and Waves, Signals and Systems, Embedded System Design.

Skills

Hardware Design and Bring-Up: Schematic capture, PCB layout review, board-level bring-up, power delivery fundamentals, signal integrity validation, component selection from datasheets

Design Tools: Cadence OrCAD, Cadence Allegro, LTSpice, SPICE simulation, analog circuit design

Test and Validation: Oscilloscopes, multimeters, logic analyzers, spectrum analyzers, signal generators, time domain reflectometer (TDR), vector network analyzer (VNA), root cause analysis, board-level debug

High-Speed and Server/Rack Architecture: AI server and GPU compute rack architecture fundamentals (coursework and independent study of platforms comparable to GB3600), high-speed digital design fundamentals, thermal and power delivery awareness

Digital Design and Firmware: Verilog HDL, RTL design, self-checking testbench development, Xilinx Vivado, Embedded C, register-level I2C/SPI/UART

Programming and Automation: Python, MATLAB, TCL, C/C++, measurement data processing

Platforms: LabVIEW, Linux/UNIX, Windows

Work Experience

Project Intern, Digital Systems Verification and Validation | Jul 2023 - Sep 2023

Advanced Systems Laboratory (ASL), Defence Research and Development Organisation (DRDO)

- Characterized fiber optic, coax, and twisted pair cable performance for a missile telemetry ground station prototype using OTDR, TDR, and VNA measurements across the S-band operating frequency range (2200 to 2290 MHz) to select the lowest-loss cable type.
- Built and validated the ground station signal chain (horn antenna, low-noise amplifier, telemetry receiver, rugged computer), reading schematics and datasheets to select and integrate components, and debugged hardware and firmware timing issues through systematic root cause analysis.
- Wrote MATLAB code to convert raw receiver voltage output into usable telemetry parameter values, turning previously unusable raw data into meaningful measurements.
- Operated oscilloscopes, multimeters, logic analyzers, and spectrum analyzers to validate signal fidelity, documenting test procedures and results in clear technical reports for repeatable use.

Projects

Pipelined Multiply-Accumulate Unit for Xilinx FPGAs | Verilog, Xilinx Vivado, Icarus Verilog

- Designed a 5-stage pipelined 32x32-to-64-bit multiply-accumulate unit in Verilog for Xilinx FPGAs, splitting a single-cycle 32x32 multiplier into four 16x16 partial products across pipeline stages to shorten the critical path.
- Authored Vivado XDC timing constraints and documented the timing-closure workflow, verified functional correctness with a self-checking testbench driving 40 back-to-back operations in Icarus Verilog.

3-Stage Class AB Power Amplifier | LTSpice, Analog Circuit Design

- Designed and simulated a high-gain (60 V/V, 35 dB) power amplifier schematic on a dual-rail +/-10V supply using TIP41C/TIP42C transistors, selecting components against a target specification, then built and validated a breadboard prototype to confirm real-world circuit behavior.
- Diagnosed crossover distortion to its root cause and resolved it via diode-based biasing, verifying clean sinusoidal output through iterative SPICE simulation and hands-on prototype testing and debug.

RISC-V RTL Implementation (5-Stage Pipelined RV32I Processor) | Verilog HDL, Icarus Verilog

- Implemented RTL for a RISC-V RV32I 5-stage pipeline with hazard detection and forwarding logic across a 12-file Verilog module hierarchy.
- Built four self-checking Verilog testbenches, debugging discrepancies to root cause and verifying all four with Icarus Verilog.

Fire Fighting Robot using Raspberry Pi | *Raspberry Pi 3B, ESP32-CAM, C/C++, Python (Team Lead)*

- Led development of an autonomous fire fighting robot on a Raspberry Pi 3B, programming Python GPIO logic to poll flame and gas sensors and automatically trigger a relay-driven water pump and buzzer alarm on detection.
- Developed ESP32-CAM firmware in C/C++ for live camera streaming and WebSocket-based remote control of dual DC motors with PWM speed control.

Solar-Based Electronic Defense Uniform | *ATmega16A, Embedded C, GPS, GSM (Team Lead)*

- Led a four-member team engineering a multifunctional embedded hardware system integrating a 5W flexible solar panel, ATmega16A microcontroller, real-time thermal management, and GPS/GSM location tracking.
- Designed power management and energy-harvesting circuitry; implemented low-level I2C/UART driver routines for peripheral integration.

Footstep Energy Harvesting System | *Piezoelectric Arrays, Power Electronics (Team Lead)*

- Led development of a piezoelectric sensor array converting mechanical energy into regulated electrical energy for mobile device charging.
- Designed signal conditioning, rectification, and voltage-regulation circuitry for analog-digital boundary circuit design.

Certifications & Achievements

- VLSI Design, Simplilearn, Issued Sep 2023, Credential ID 2176287
- Digital Signal Processing, Simplilearn, Issued Dec 2023, Credential ID 2176283
- Embedded System Design, Simplilearn, Issued Feb 2024, Credential ID 2176281
- Project Certification: Simulation and Measurement of Telemetry Receiving System, DRDO
- PCB Design: Cadence OrCAD and Cadence Allegro Expert Certification, Udemy
- FPGA Engineer: Verilog with Xilinx Vivado Design Suite, Udemy
- Verilog HDL: VLSI Hardware Design Comprehensive Masterclass, Udemy